

Two years with Turing.jl v0

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The Alan Turing Institute, UK (*until recently*)

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I only have ≤ 15 minutes!

I only have ≤ 15 minutes!

```
julia> using Dates
```

```
julia> 2 * Day(365) / Minute(15)  
70080.0
```


I only have ≤ 15 minutes!

Overview

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Overview

3 good things

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3 good things

2 things to improve

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N questions

Two years with Turing.jl v0

Overview

Turing provides a macro-based DSL to express probabilistic models

3 good things

2 things to improve

N questions

```
@model function eight_schools(y, sigma)
    mu ~ Normal(0, 5)
    tau ~ truncated(Cauchy(0, 5); lower=0)
    theta ~ MvNormal(fill(mu, length(y)), tau^2 * I)
    for i in eachindex(y)
        y[i] ~ Normal(theta[i], sigma[i])
    end
end
```


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Overview

Turing provides a macro-based DSL to express probabilistic models

3 good things

and (mostly Bayesian) inference algorithms

2 things to improve

```
@model function eight_schools(y, sigma) ... end
```

```
y = [28, 8, -3, 7, -1, 1, 18, 12]  
sigma = [15, 10, 16, 11, 9, 11, 10, 18]
```

N questions

```
model = eight_schools(y, sigma)  
sample(model, NUTS(), 1000)
```

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I & others worked on it July 2024 – May 2026

N questions



penelope 25 Jun at 5:06 PM

the use of filldist, which does not give pointwise loglikelihoods.

you might be able to pass `factorize=true` to loo to make it do so: <https://pysm.dev/FlexiChains.jl/stable/integrations/#PosteriorStats.loo-Tuple{DynamicPPL.Model,%20FlexiChains.FlexiChain}>

since when does this work when the LHS is an Array?

Since forever, basically. I don't like it myself github.com/TuringLang/DynamicPPL.jl/issues/919
It only "works" for observations, and not random parameters. (edited)

← #turing

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But I'm a software engineer, *not* a statistician!

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Overview

(1) Modularity

3 good things

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(*) simplified for
purposes of this talk

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(1) Modularity

State type used for model evaluation:

```
struct VarInfo{V<:Dict,L<:Real}  
    values::V  
    logp::L  
    # other fields ...  
end
```

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Can't be extended, can't be trimmed down!

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end
```

Can't be extended, can't be trimmed down!

`values` used as both input and output (!!)

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(1) Modularity

New accumulators represent outputs

```
struct VarInfo{A<:Tuple{Vararg{Accumulator}}}  
    accs :: A  
end
```


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(1) Modularity

New accumulators represent outputs

```
struct VarInfo{A<:Tuple{Vararg{Accumulator}}}  
    accs :: A  
end  
  
struct LogLikeAcc{L<:Real} <: Accumulator  
    loglike :: L  
end  
  
struct LogPriorAcc{L<:Real} <: Accumulator  
    logprior :: L  
end
```

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(1) Modularity

New accumulators represent outputs

New “initialisation strategies” represent inputs

Init strategy → **Model** → Accumulators

Custom subtypes give Gen-like flexibility

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(2) Performance

Drastic reduction in Turing's overheads (>1000x in some cases)

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(2) Performance

Drastic reduction in Turing's overheads (>1000x in some cases)

Mostly comes from judicious use of accumulators, but also new data structures ([VarNamedTuple](#))

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(2) Performance

Drastic reduction in Turing's overheads (>1000x in some cases)

Mostly comes from judicious use of accumulators, but also new data structures (`VarNamedTuple`)

With reverse-mode Enzyme, faster than Stan across various models

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(3) Fidelity

Single type for mapping variable names => values

VarNamedTuple

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(3) Fidelity

Single type for mapping variable names => values

`rand(model)`



VarNamedTuple



`maximum_likelihood(model)`

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(3) Fidelity

Single type for mapping variable names => values

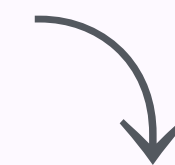
`rand(model)`



VarNamedTuple



`maximum_likelihood(model)`



`sample(model...;
initial_params=...)`

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3 good things

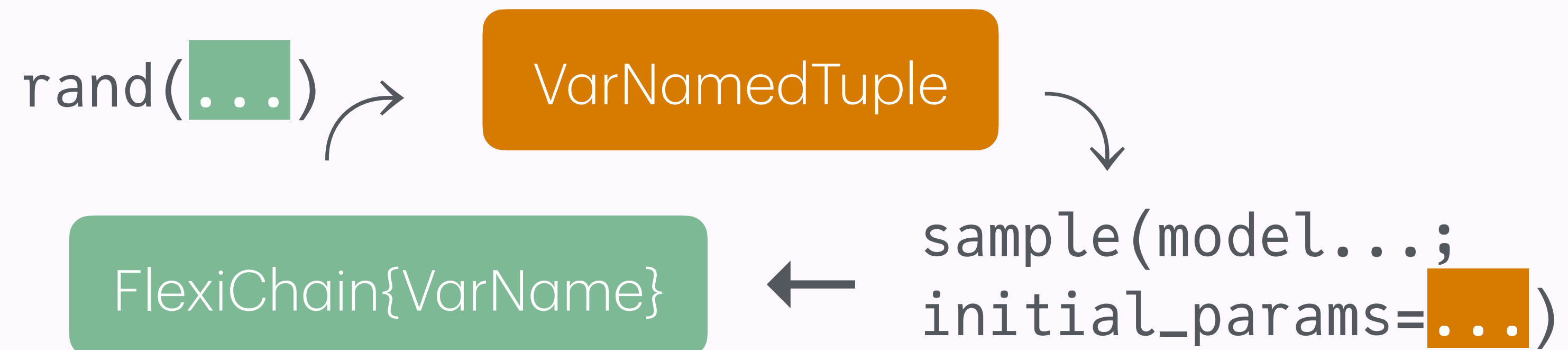
2 things to improve

N questions

(3) Fidelity

Single type for mapping variable names => values

Chain storage that preserves shapes and types



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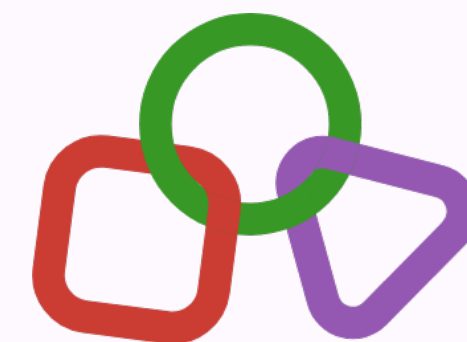
(3) Fidelity

Single type for mapping variable names => values

Chain storage that preserves shapes and types

FlexiChains.jl isn't just for Turing – get in touch :)

FlexiChain{VarName}



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(1) Identifying users

I'm a software engineer, *not* a statistician...

```
@model function f(y)
    x ~ Normal()
    y ~ Normal(x)
end
```

My favourite model!

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(1) Identifying users

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Need to reach out to users to understand what pain points exist

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(1) Identifying users

I'm a software engineer, *not* a statistician...

Need to reach out to users to understand what pain points exist

Challenge ourselves to do *better*!

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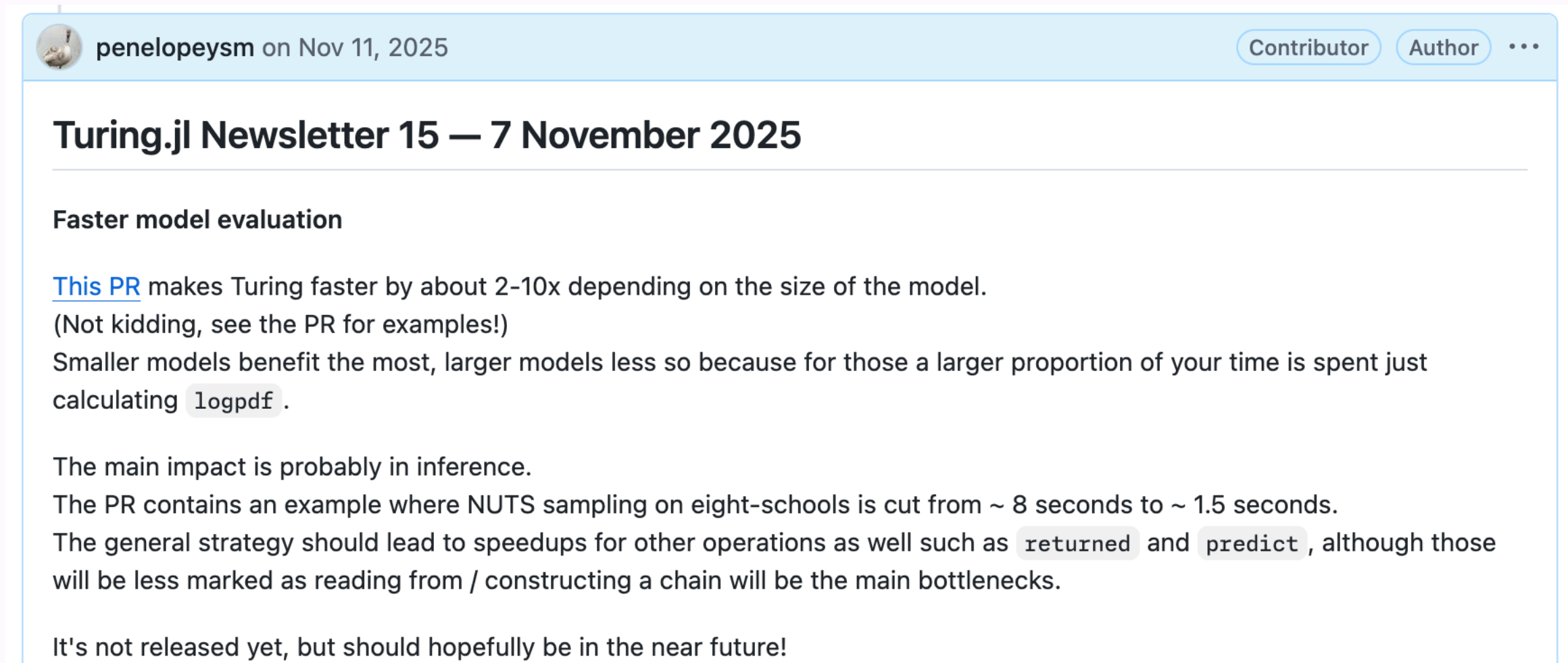
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(2) Outreach beyond Julia

Newsletters on GitHub, Julia Slack, website



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Newsletters on GitHub, Julia Slack, website

Dev team

Julia users
(maybe using Turing)

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(2) Outreach beyond Julia

Newsletters on GitHub, Julia Slack, website

Dev team

Julia users
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The world (incl. users of Stan,
Numpyro, etc...)

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Thanks to:

- My former colleagues on TuringLang
- Everyone who's engaged with us on GitHub, Slack, and beyond — it's been the highlight of my time on this
- You!

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Overview

Slides + a longer written version of this talk

<https://pysm.dev/talk>

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